

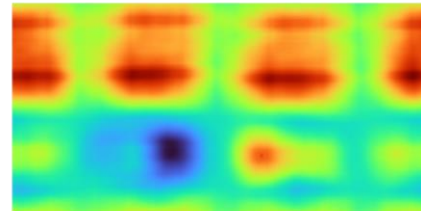
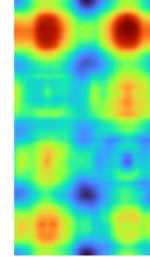
Initial Image



Element to be Searched



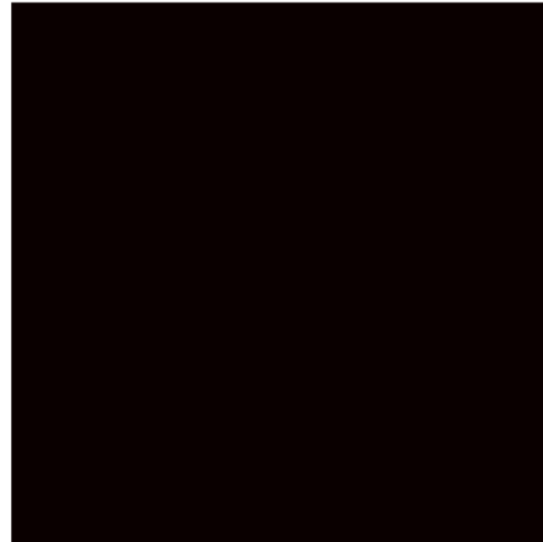
Heat Map



Aperture



Laser Diffraction Pattern



# Fourier Transform Model of Image Formation (I of II)

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# Objectives

- To determine the Fraunhofer diffraction pattern that will be produced by different apertures when exposed to monochromatic light using a Fast Fourier Transform (FFT) algorithm .
- To simulate a cataract by determining the image that will produced by different apertures when exposed to an arbitrary image using convolution.
- To use an image correlation algorithm to determine the locations of details in an image

# Fast Fourier Transform (FFT)

The Fast Fourier Transform Algorithm is a trick discovered by Carl Friedrich Gauss to calculate a Discrete Fourier Transform (DFT) in a very short time  $O(N \log N)$  [1]. Obviously he wasn't able to utilize his algorithm in its maximum potential, but thankfully we are graced with the ability to tell a computer to perform tedious calculations at lightning speed and without error.

According to theory [2], if you take the FT of a function once you will get its frequency distribution in the Fourier space. If you then take the FT of the function once more, you will get the original function but with a sign change in the independent variable. If you decide to take the FT four consecutive times however, you will get the initial function. Obviously, a much more efficient way to retrieve the original function is to take the FT and then take the inverse FT. This will transport the function into the Fourier space, then back again to the real space. Of course all of these results will translate to a 2D function as well.

Hence, **if you want to recover the initial image from the FT you can 1) take the inverse FT or 2) take the FT twice again.**

Image



Fourier Transform Twice



Fourier Transform then Inverse



# Fraunhofer Diffraction

The Fraunhofer diffraction condition is a model used to describe plane waves incident on an aperture which is then viewed on a screen very far away. The diffraction patterns can be solved analytically by the following equations:

Slits [3] :

$$a \sin \theta = m\lambda$$

which is the ever-familiar diffraction equation.

Circular Aperture [4] :

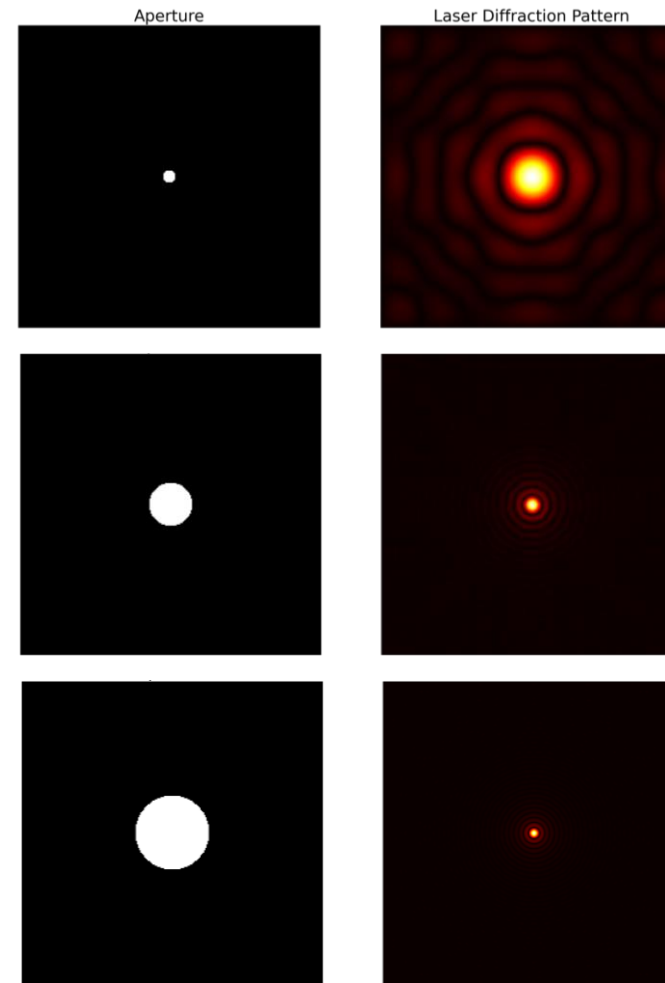
$$I(\rho) \propto \left( \frac{2J_1(\pi\rho D/\lambda z)}{\pi\rho D/\lambda z} \right)^2$$

where

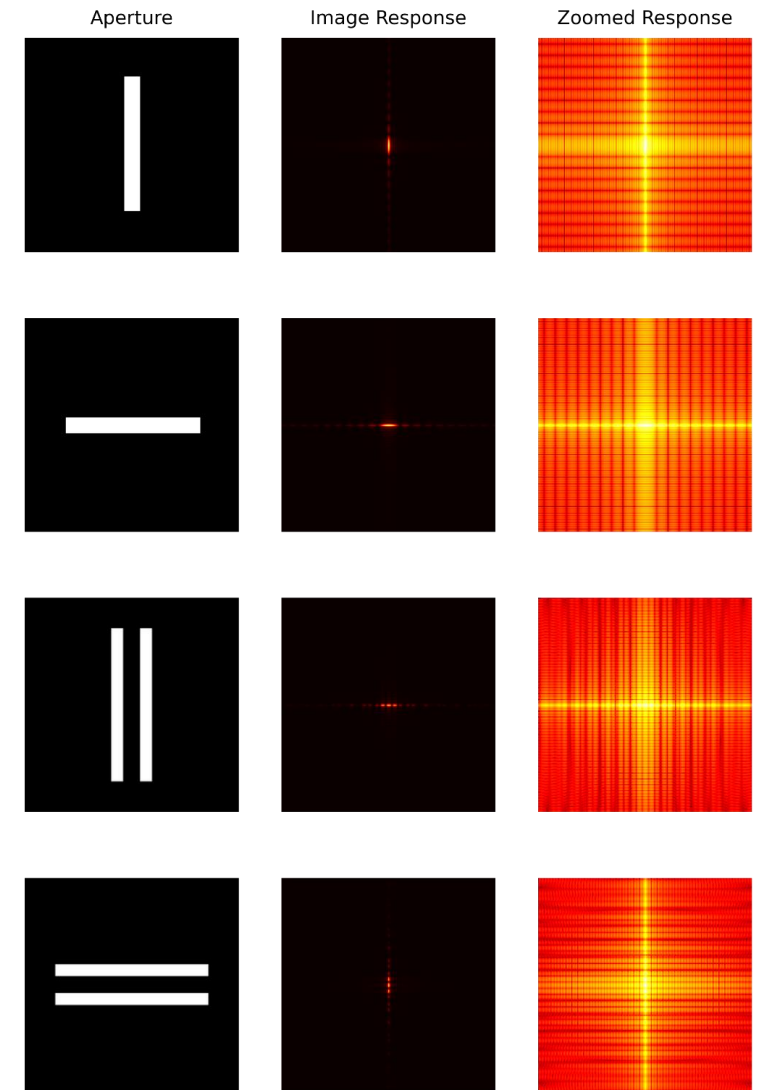
$\rho$  = radial distance from the optical axis

$D$  = aperture diameter

$J_1$  = First Bessel Function



**Fig 1.** Circular Aperture Diffraction Pattern Snapshots

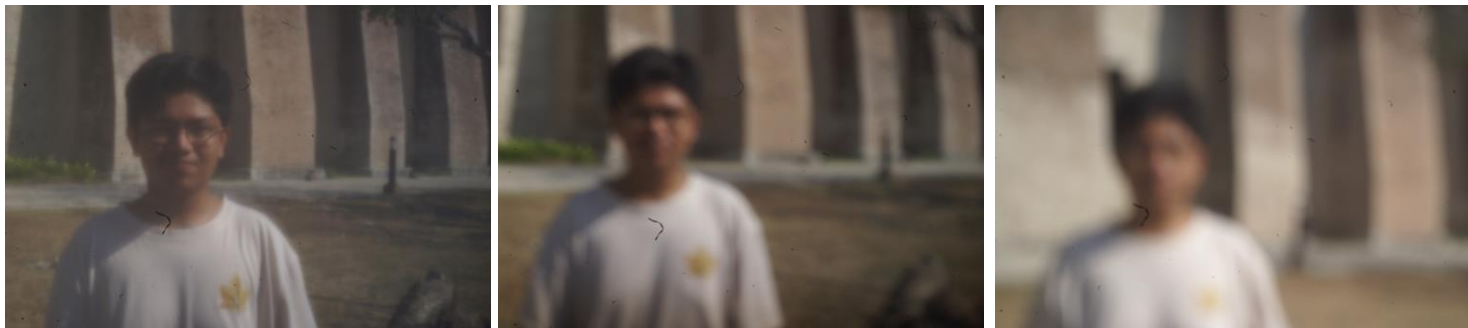


**Fig 2.** Diffraction Patterns Produced by Slits

# Fraunhofer Diffraction

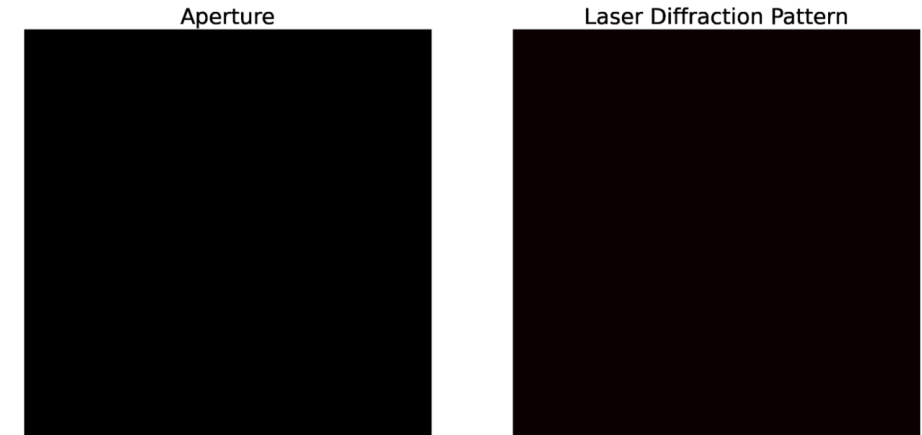
In order to simulate the diffraction patterns, the Fast Fourier Transforms (FFT) of the binarized aperture images were shifted and their absolute values taken in order to produce a coherent image. **We can see from Fig 2. that the diffraction patterns from the simulated slits agree with the equation and results from laboratory experiments.** The slits limit the amount of wavefronts that can enter through the barrier, and so the resultant wave converts from a plane wave to a spherical wave – by Huygen's Principle.

**For a circular aperture (Fig 1. and Fig 3.), the same principle applies except that the diffraction pattern produced will be radially symmetrical.** As the diameter of the pinhole is increased, the diffraction pattern becomes less pronounced since more wavefronts are let through – minimizing spherical wave conversion.



**Fig 4.** Circular Aperture with Varying Size – Sharpness Decreases with Increasing Pinhole Size [5]

Link to GIF : <https://imgur.com/a/HS90sml>



**Fig 3.** Circular Aperture with Varying Size (GIF)

We can use the results from these exercise to predict a possible anomaly in the images produced by a pinhole camera. A small pinhole is expected to produce a sharper image since the projection would practically be as small as the pinhole (or even smaller) as seen in Fig 4. However, if the pinhole is extremely small, the image will paradoxically become blurry due to the diffraction we see in Fig 3. **Hence, a camera obscura will produce a sharper image given a smaller pinhole but ONLY up to a certain point.**

# Convolution

The convolution of two functions is given by the integral [6]

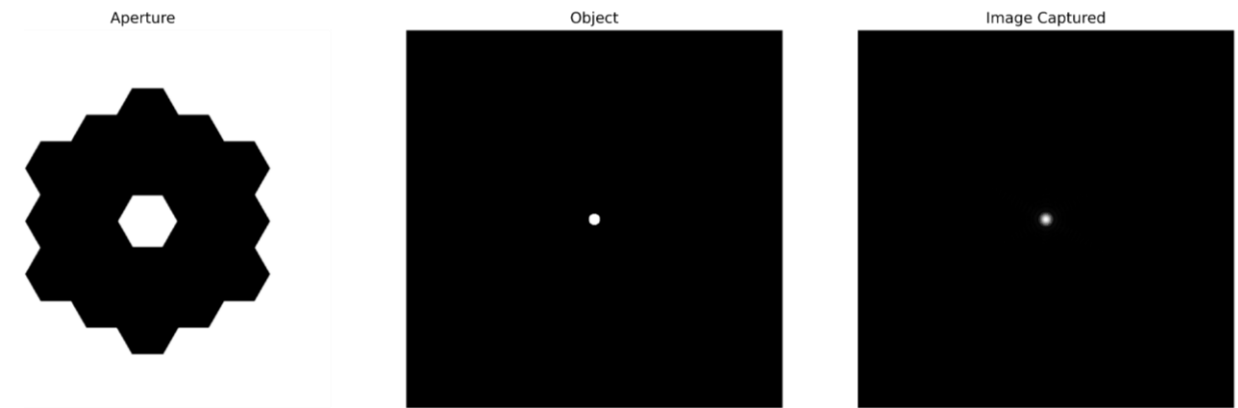
$$h(x, y) = f \circledast g = \iint f(x', y') g(x - x', y - y') dx' dy'$$

which is a challenge to evaluate for a lot of functions. Alternatively we can take the FT of  $f$  and  $g$  since this operation becomes much simpler in the Fourier domain. We have

$$H = FG$$

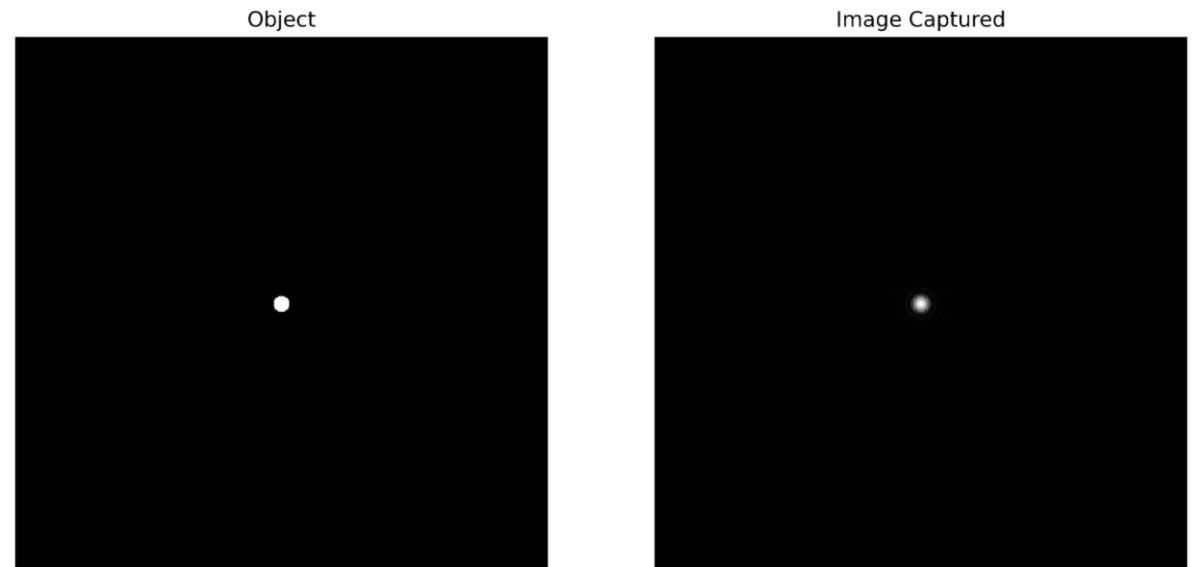
which is now just a simple product. All that's left to do is to go back to real space by applying an IFT. In a nutshell, this operator “smears” two functions together such that the result looks similar both the operands.

**In this exercise, we multiply an aperture to the FT of a desired object to get the image produced by the imaging system.** In this case, the aperture is already considered to be in the Fourier space, while the object is in the real space.



**Fig 5.** A Distant Star Captured by the James Webb Space Telescope

**In the case of the JWST, the resulting image was smoother than the original, although it had circular patterns near the edge of the object.** We can see it more clearly below:



# Convolution

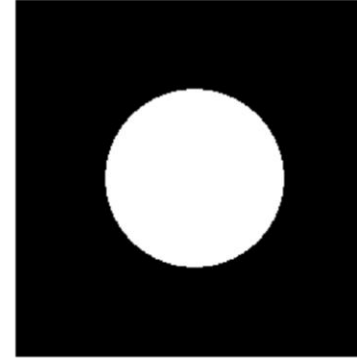
In the previous slide, we determine the image that will be captured by the JWST given that the object is a distant star. **Here, we try to simulate what a cataract will do to a human's vision.** We do this by simulating an aperture with an annulus in the middle as shown in Fig 6. In the figure, **a circular aperture seems like an appropriate approximation for the human eye** due to the fact that the image appears to have no obstructions. It's not a perfect approximation since the image has some slight perturbation in its interior, but the general appearance is correct.

Shown in the second column is the same aperture but with an annulus of significant size. The image produced in this case is very tainted and we can only see the edges of the object, along with a “diffraction pattern” at the exterior. **We can immediately see that an annulus/cataract can be used for edge tracing an image** – it is rather easy to implement so it can act as a preliminary tool before using other sophisticated methods.

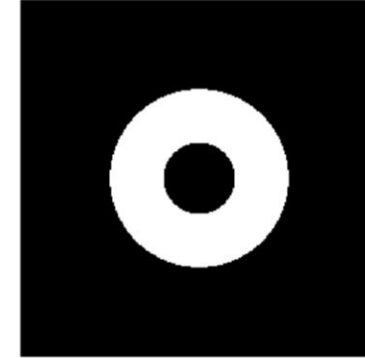
**If we reduce the size of the “cataract” we see that the diffraction patterns become more evident. This is due the phenomena shown in Fig 1.** A small aperture will yield a diffraction pattern once a Fourier operator is applied to it. This means that even if there was no cataract, a small aperture would still yield these patterns.

## Cataract Simulation using Convolution

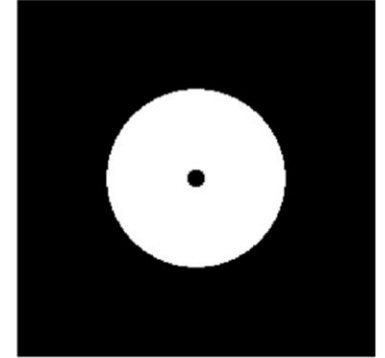
Human Eye Approximation



Cataract Approximation



Small Cataract



Human Eye Result



Cataract Result



Small Cataract Result



**Fig 6.** Simulating a Cataract in a Grayscale Setting



# Correlation

The correlation of two functions is given by the integral [6]

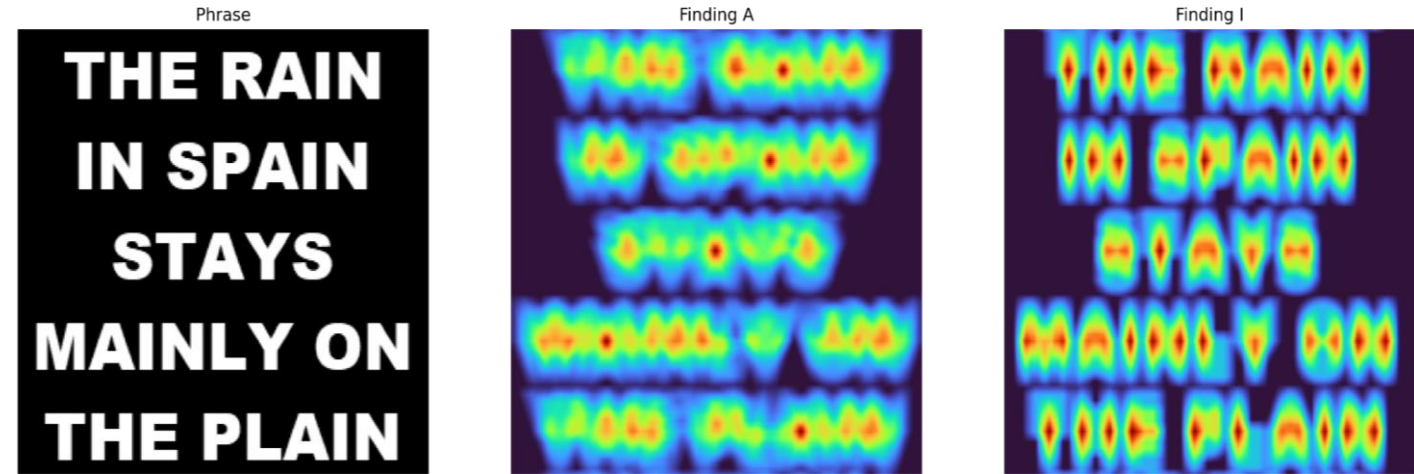
$$p = f \odot g = \iint f(x', y') g(x + x', y + y') dx' dy'.$$

which is hard to evaluate for a lot of functions. We have a simpler relation in the Fourier space:

$$P = F^* G$$

where  $*$  denotes a complex conjugate. Like in convolution, we use the Fourier space to do the math then convert back to real space to display results. In a nutshell, this operator measures the similarity of two functions. A higher similarity in a certain position between the functions means a higher correlation value.

**In this exercise we develop code that will correlate two images – this can be used to find the locations of image details based on the resulting correlation map (like finding Waldo, for example).**



**Fig 7.** Finding the Symbols “A” and “I” in a Sentence

In Fig 7., we use image correlation to find the letters “A” and “I” in the above sentence. In the formula on the right  $F$  stands for an image of these letters. **We can see that the correlation peaks with the letter “A” are well defined and correspond precisely to its locations. On the other hand, the correlation peaks with the letter “I” are all over the place. This is because the former has a very distinct shape that no other letter possesses while an “I” pattern can be found on almost any letter.**

The correlation operator can then be used to find objects in an image, although it can only detect general shapes and appearances – after all, it’s just simple multiplication.



# Correlation

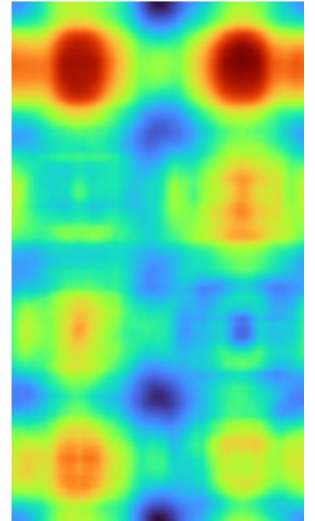
Initial Image



Element to be Searched



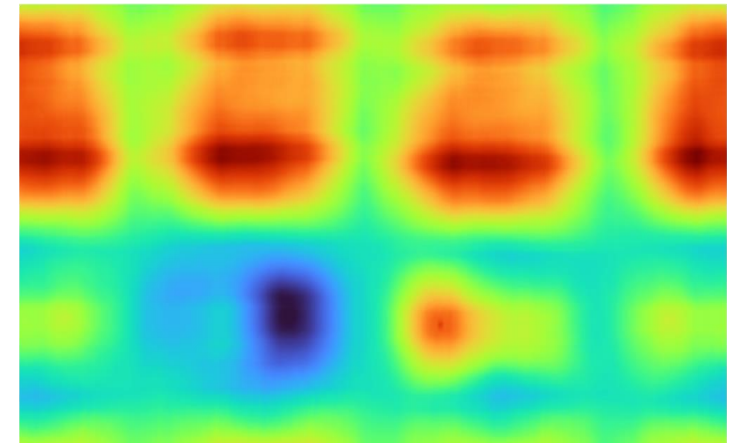
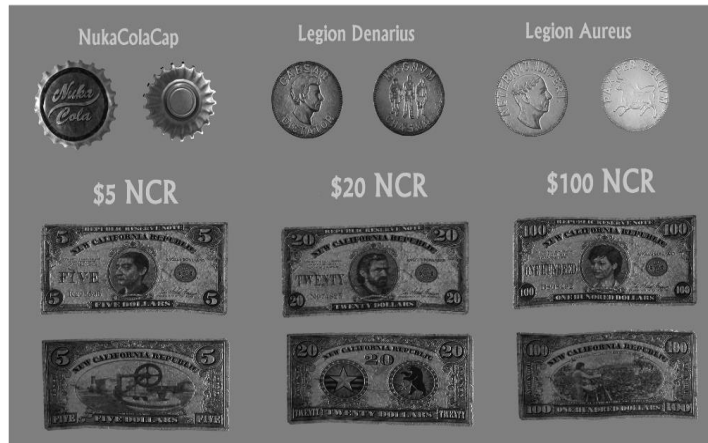
Heat Map



**Fig 8.** Finding the Caesar's Legion Flag [\[7\]](#)

In the previous slide, we used correlation to find certain letters in a string of text. Here, we use it to find objects in non-binarized images. First we focus on the top part, which is an image that contains the insignias of major factions in the Mojave wasteland (as you can tell, I am really a big fan of Fallout). **The element to be searched for is the flag of Caesar's Legion which is a Roman-era inspired bull symbol.** Using image correlation, a heat map was generated which showed two peaks. **This result is interesting because both peaks correspond to flags that shows an animal** (a bull for Caesar's Legion and a two-headed bear for the NCR). All other symbols such as the skull and the crown had low correlation values. **This algorithm is not perfect but it's definitely not useless – now I only have to look at two symbols instead of eight in order to confirm which flag I want!**

# Correlation



**Fig 9.** Finding the Nuka Cola Bottle Cap (Bottom) [7]

Shown in this slide are various currencies used in the Mojave wasteland, the most stable of which is the Nuka Cola Bottle Cap. We try to find the front-facing bottle cap to test the limits of image correlation in determining the details on an object. Based on the generated heat map, the correlation values are high on both Nuka Cola Caps and Legion Denarius/Aureus. The algorithm wasn't able to differentiate the engraving between these coin currencies, but it was able to differentiate it from the paper money of the New California Republic. **In other words, it was only able to differentiate between coins and paper – not the details of the coins itself.** This is to be expected since we are only working on a multiplication in Fourier space. If we want this level of precision, machine learning should be the approach!

# Key Takeaways

- The Fast FT is very useful in taking the FT of discrete data points – especially when the number of points is  $2^n$
- The FT of an aperture's image is its response to an incoming monochromatic plane wave.
- The convolution of two objects yields a mixture of the two and can be used to simulate an object's image in an aperture
- The correlation of two objects yields a correlation map that yields the position of their similarities.

# Reflection

I enjoyed this module much more than the first one. Maybe it was due to the learning curve or maybe because the results in this module were a bit more suitable for my taste – but the things I learned here will probably stick to me more. This was also my first proper introduction to the convolution and correlation operators which was useful since they are terms that the people in my laboratory frequently use! Now I know what it means! I exerted more effort in this module than my first, and I think I made enough data sets and enough suitable conclusions. **Due to this I give myself a score of 9.5 – 10 /10.**

# References

- [1] Reyes, R. Fourier Transform – Module 3. (2022). Applied Physics 155.
- [2]<https://dsp.stackexchange.com/questions/24880/fourier-transform-4-times-original-function-from-bracewell-book>
- [3] <http://hyperphysics.phy-astr.gsu.edu/hbase/phyopt/fraungeo.html>
- [4]<https://www.opticsthewebsite.com/Circular#:~:text=The%20Fraunhofer%20diffraction%20pattern%20of,is%20the%20First%20Bessel%20Function.>
- [5] Soriano, E., Ducusin M., and Loot A. (2023). Physics 165 Pinhole Camera Experiment.
- [6] Soriano, M. (2023). Activity 2. Fourier Transform Model of Image Formation (1 of 2)
- [7] Obsidian Studios. Fallout: New Vegas Game Files.